

WELL INTERVENTION
IN PRACTICE

Engineering Decisions, Well Control & Operational Lessons

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The Well Is Already Producing

Why Do We Need to Go Back In?

WHY WE GO BACK

- PERFORMANCE
- INTEGRITY
- COMPLETION
- RESERVOIR
- DIAGNOSTICS

A producing well is not a static system

Performance, integrity and downhole conditions evolve over the well life



Where Does Intervention Fit?

Intervention can recur throughout the producing life of a well

PLAN › DRILL › COMPLETE › PRODUCING LIFE › ABANDON

MAY RECUR THROUGH WELL LIFE

PRODUCE ⇄ INTERVENE

A producing well is not a static system

Performance • Integrity • Completion • Reservoir • Surveillance

KEY POINT End of drilling ≠ End of downhole challenges



What Are We Trying to Achieve?

Define the objective before selecting the intervention method

FIVE COMMON OBJECTIVES — NOT A FIXED SEQUENCE

01	**DIAGNOSE**	Understand what is happening
02	**MAINTAIN**	Preserve current well capability
03	**RESTORE**	Recover capability that has been lost
04	**REPAIR**	Correct a mechanical or integrity problem
05	**IMPROVE**	Enhance performance beyond the current condition

KEY MESSAGE **Objective first. Method second.**



DO NOT START WITH THE TOOL

Tool availability is a constraint — not the starting point

TOOL-FIRST THINKING

- CT is available → use CT?
- Run slickline first and see?
- Previous well used the same method?

START WITH

THE WELL PROBLEM

PROBLEM → EVIDENCE → OBJECTIVE → METHOD

Availability influences the solution

It should not define the problem

Before method selection, confirm barriers, risks, access and operating constraints



Diagnose the well condition before choosing how to intervene

What We See May Only Be a Symptom

One observed change can have several plausible causes

- Scale / deposits?
- Tubing / completion restriction?
- Artificial-lift performance?
- Surface / choke constraint?

OBSERVED SYMPTOM

900 → 300

BOPD

- Formation damage?
- Reservoir pressure decline?
- Water / gas breakthrough?
- Mechanical / integrity issue?

SYMPTOM ≠ DIAGNOSIS

First compare the data on a like-for-like operating basis



Start With the Well History

Reconstruct the sequence of events before deciding how to intervene

01	**WHAT CHANGED?**	Rate, pressure, WC/GOR, lift behavior, annulus or integrity signal?
02	**WHEN DID IT CHANGE?**	Fix the timing, then relate it to operating or intervention events
03	**SUDDEN OR GRADUAL?**	A step change and a slow decline usually imply different mechanisms
04	**WHAT HAPPENED JUST BEFORE?**	Start-up/shutdown, choke or lift change, pumping, workover, facility upset?


History narrows hypotheses — verification confirms

Reconstruct the chronology before selecting the method

What Information Do We Need?

Build the intervention basis from verified data — not assumptions



WELL & COMPLETION

- Completion / wellhead schematic
- Trajectory, depths & access path
- Minimum ID, restrictions & barrier status

PERFORMANCE & SURVEILLANCE

- Oil / gas / water rate history
- Flowing & static pressure history
- Artificial-lift & operating data

INTEGRITY & PRIOR WORK

- Tubing / casing / annulus integrity
- Barrier / leak-test history
- Previous interventions / workovers

FLUID & RESERVOIR

- PVT / fluid properties & compatibility
- Reservoir pressure, PI / skin
- Contacts, zonal contribution & depletion

Verified data define the intervention basis; objective, barriers, risk and constraints shape the program

Surface Indication Does Not Equal Root Cause

The first indication is a clue — not a conclusion

? **HIGH TUBING PRESSURE** **does not identify one unique cause**

Downstream backpressure? choke? restriction? fluid column? lift behavior?

? **LOW PRODUCTION** **does not automatically mean scale**

Reservoir/inflow? lift? tubing restriction? breakthrough? surface constraint?

? **ANNULUS PRESSURE** **is an integrity indication — not proof of a tubing leak.**

Trapped/thermal pressure? packer? tubing? casing? communication?



OPERATING DISCIPLINE

INDICATION ≠ DIAGNOSIS

Test the hypothesis against history, trends, well configuration, barrier status and targeted diagnostics

Would You Approve an Intervention Yet?

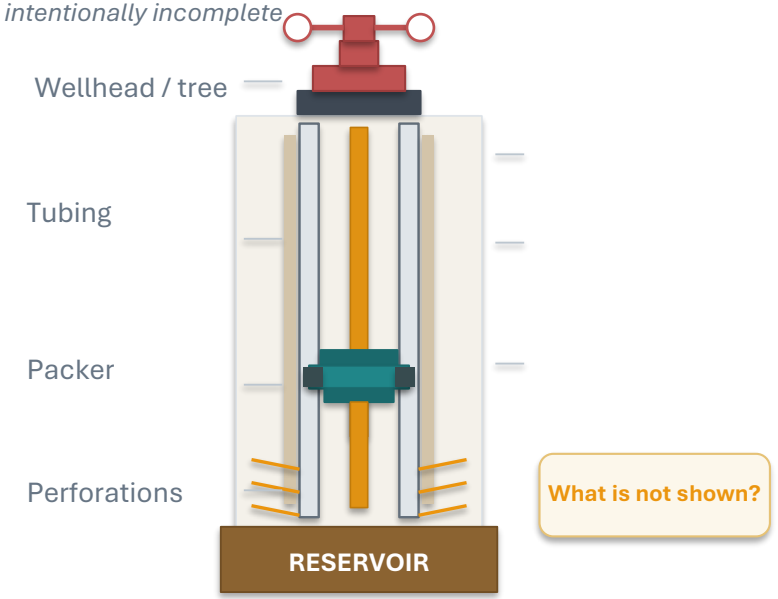
CASE 01

5 MIN

You have a simplified completion sketch and an oil-rate decline. What information is still missing?

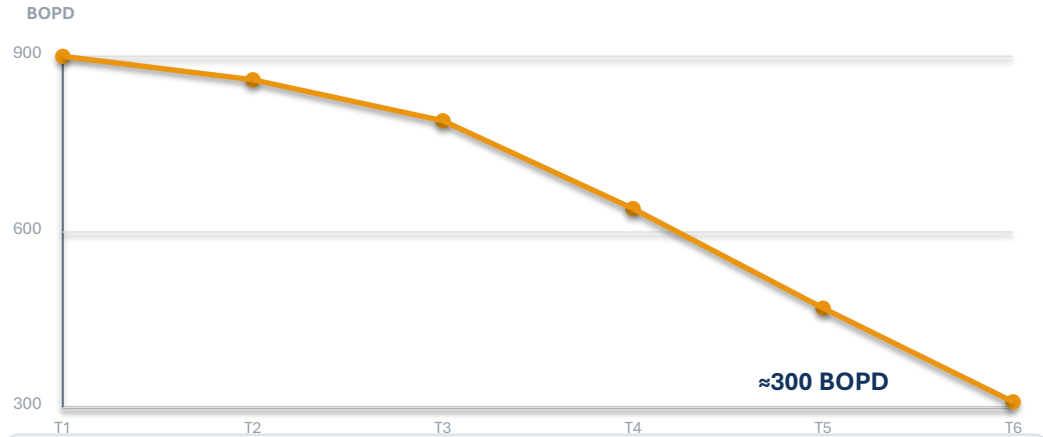
WHAT YOU KNOW — COMPLETION

Simplified and intentionally incomplete



WHAT YOU KNOW — PERFORMANCE

Illustrative oil-rate trend; operating context not shown



Comparable test basis not confirmed

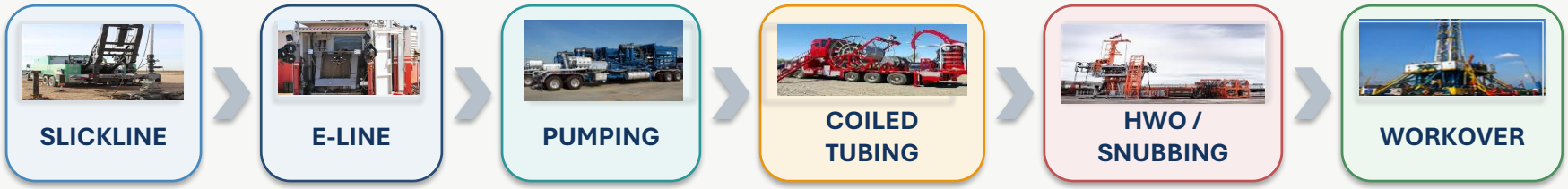
What would you ask before approving the intervention objective and method?

Identify the minimum additional information you need — not every possible data item

From Light Intervention to Workover

Well intervention is a spectrum — workover sits within that spectrum

INTERVENTION SPECTRUM



Boundaries overlap; classification varies with job scope, live-well status and equipment

WORKOVER IS WELL INTERVENTION

Usually the heavier / rig-based end of the intervention spectrum

METHOD ≠ OBJECTIVE

Choose the method only after the well problem and objective are clear

The method should fit the well — not the well fit the available method







What Changes Across the Spectrum?

Capability, access, pressure-control needs and operating footprint change together



SLICKLINE



E-LINE



PUMPING



COILED TUBING



HWO / SNUBBING



WORKOVER

MECHANICAL CAPABILITY	light shifting / setting	logging / conveyed tools	fluid placement	circulation / cleanout	rigid tubular work under pressure	pull / rerun completion
CIRCULATION / FLUID ACCESS	limited	limited	surface pumping path	strong	job-dependent	strong / full workover system
ACCESS / CONVEYANCE	wire	electric cable	existing flow path	continuous tubing	jointed pipe / hydraulic unit	rig / work string
PRESSURE-CONTROL COMPLEXITY	live-well PCE	live-well PCE	pump / well barriers	CT PCE + returns	high, live-well stack	full workover well-control system
FOOTPRINT / TIME / COST	low	low–moderate	low–moderate	moderate	high	high

This is a capability / complexity spectrum — not a risk ranking. Every method still requires an appropriate barrier and well-control plan.







Slickline & E-Line — When Are They Enough?

Excellent for access, diagnostics and selected mechanical tasks — when circulation or heavy mechanical work is not required

SLICKLINE



WIRE / MECHANICAL

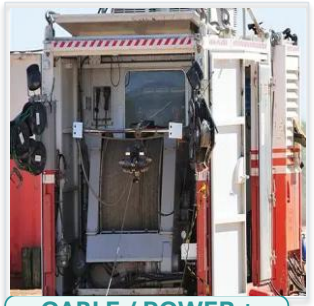
Typical strengths

- Set / retrieve selected flow-control devices
- Shift sleeves and operate simple downhole mechanisms
- Gauge / memory data runs
- Selected fishing and mechanical manipulation

Watch-outs

- Live-well pressure-control setup
- Minimum ID / deviation / toolstring reach

E-LINE / WIRELINE



CABLE / POWER + DATA

Typical strengths

- Depth correlation, logging and production diagnostics
- PLT / temperature / noise / integrity diagnostics
- Perforating or setting tools where the system is designed for it
- Tractor / conveyance options in selected deviated wells

Watch-outs

- Cable tension / reach / conveyance
- Pressure-control and explosive / radiation controls where applicable

If the objective requires significant circulation, cleanout, weight/torque or major completion work, move to another method







Pumping & Coiled Tubing — When Fluids and Circulation Matter

The key distinction is whether the job only needs fluid placement, or also needs controlled downhole conveyance and circulation

PUMPING



Use when the objective can be achieved through an established flow path

- Kill / displacement fluids where planned
- Solvent / scale / chemical treatment
- Matrix stimulation or fluid placement where injectivity allows
- Pressure test / function support as applicable

pump / fluid source — **WELL / FLOW PATH** — **fluid placed at target**

COILED TUBING



Use when the job needs controlled conveyance, circulation or localized treatment

- Circulate / clean out deposits and fill
- Spot fluids accurately at depth
- Nitrogen lift / unloading and selected stimulation work
- Convey tools or diagnostics in selected applications

CT reel / injector — **DOWNHOLE ACCESS** — **circulate / convey / spot treatment**

CHECK BEFORE SELECTING: minimum ID • depth / deviation • circulation path • returns handling • pressure limits • CT fatigue / reach • barrier & PCE arrangement





Hydraulic Workover / Snubbing

Rigid tubular conveyance under pressure can extend capability beyond CT — but pressure-control and force management become central

FIELD CONFIGURATION



WHAT THE SYSTEM MUST CONTROL

HYDRAULIC FORCE CONTROL
jacks / pipe force

PCE / BARRIER STACK
stripper • rams • BOP

WELLHEAD / LIVE WELL
pressure interface

Concept only — actual unit, PCE stack and barrier philosophy are job-specific.

WHEN IT MAY FIT

- Run / pull rigid tubulars with the well live, where the approved plan requires it
- Completion or remedial work needing greater mechanical capability than wireline / CT
- Applications where killing the well is undesirable or incompatible with the objective

OPERATING DISCIPLINE

- Barrier / pressure-control stack and verification
- Pipe-light / pipe-heavy force balance and handling
- Tubular buckling / movement / depth control
- Detailed well-control procedures, contingencies and competent crew







Workover — When Major Rig-Based Intervention Is Required

Workover is part of well intervention, typically used when the completion or major downhole equipment must be pulled, repaired or changed



TYPICAL WORKOVER SCOPE

- Pull / rerun tubing or completion string
- Packer replacement or major completion repair
- Recompletion / zonal isolation / remedial cementing
- Major integrity repair or artificial-lift completion change
- Fishing / remedial work requiring rig-based handling

WHAT CHANGES OPERATIONALLY

- Larger footprint, tubular handling and logistics
- Full workover program, BOP / barrier verification and contingency planning
- Potential well kill / displacement depending on well state and approved strategy

Organizational labels may separate “Intervention” and “Workover”; engineering-wise, workover remains within the intervention spectrum







Choose the Method That Fits the Well

The objective sets the direction; barriers, well condition and operating constraints determine what is feasible

1



OBJECTIVE

What must be achieved?

2



EVIDENCE

What does the data support?

3



BARRIERS & WELL CONTROL

Can we maintain the required barrier envelope?

4



ACCESS & GEOMETRY

Can the method reach and operate were needed?

5



CAPABILITY

Can it actually execute the task?

6



RISK / LOGISTICS / CONTINGENCY

Can it be executed reliably and recoverably?

SELECTION QUESTION

Can this method safely and reliably achieve the objective within the well's actual constraints?

Do not choose the lightest method by habit — and do not choose the heaviest method for comfort

Choose the method that safely, reliably and efficiently achieves the objective



Before We Enter the Well, Ask: What Holds the Pressure?

Every intervention method changes how we access, verify or temporarily disturb the well's pressure-control envelope

INTERVENTION CONTROL LOGIC



SOURCE OF PRESSURE

reservoir / well pressure



BARRIER ENVELOPE

verified pressure barriers



PRESSURE-CONTROL SYSTEM

PCE / stack / containment



CONTROLLED ACCESS

intervention through PCE

THREE QUESTIONS

- What is the current barrier envelope?
- What changes when the intervention equipment is installed or operated?
- How will pressure be contained if the planned barrier is lost?

Pressure control is not an equipment list — it is a barrier strategy



A Barrier Is More Than a Valve

A credible barrier depends on defined elements, acceptance criteria, verification and known status

CONCEPTUAL BARRIER ENVELOPES



SOURCE OF INFLOW



PRIMARY BARRIER ENVELOPE

First verified set of barrier elements preventing flow from the source.

Status must be known



SECONDARY BARRIER ENVELOPE

Independent back-up barrier where required by the well state and governing standard

Independence matters



PEOPLE / ENVIRONMENT

WHEN CAN WE CREDIT A BARRIER?



DEFINED

Which elements make up the envelope?



ACCEPTED

Are acceptance criteria met?



VERIFIED

Has status / function / pressure integrity been verified as required?



MONITORED

Can degradation or loss be detected?

During intervention, re-check the barrier envelope whenever equipment, well status or flow path changes

M4



Live Well or Killed Well?

The decision is a well-control strategy — not a preference for one intervention method

LIVE-WELL STRATEGY

Well pressure remains present and is controlled through the defined barrier / PCE system



WHY IT MAY BE SELECTED

- Avoid unnecessary kill-fluid exposure, losses or formation impairment
- Maintain access to a naturally pressured well where the approved plan supports it
- Enable selected wireline, CT or snubbing operations under controlled pressure

reservoir pressure — **PCE + BARRIERS** — **controlled access**

KEY DISCIPLINE

Pressure-control equipment and verified barriers remain active throughout the exposure

KILLED-WELL STRATEGY

A planned hydrostatic / fluid barrier controls the well as part of the approved barrier envelope



WHY IT MAY BE SELECTED

- Provide a stable hydrostatic barrier for selected workover / completion tasks
- Simplify some tubular or completion operations where appropriate
- Support rig-based work when the well can be safely displaced and maintained

fluid column / MW — **HYDROSTATIC BARRIER** — **controlled well state**

KEY DISCIPLINE

Killed does not mean zero well-control risk; losses, influx and barrier degradation remain possible

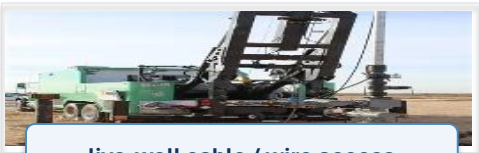
Choose the well-state strategy from pressure, reservoir response, integrity, objective, barrier feasibility and contingency — not convenience



Pressure-Control Equipment Must Match the Method

The stack is not generic. It must be compatible with the conveyance, well state, pressure, toolstring and contingency plan.

WIRELINE



- live-well cable / wire access
- Lubricator / pressure-control head
- Wireline valve / BOP as applicable
- Grease injection / stuffing box depending line type

Concept only.

Actual configuration is job-specific.

COILED TUBING



- continuous tubular + returns
- Injector / stripper
- CT BOP system
- Flow / returns control and contingency

Concept only.

Actual configuration is job-specific.

HWO / SNUBBING



- rigid tubular under pressure
- Stripper / seal system
- Ram / BOP configuration
- Hydraulic unit and force control

Concept only.

Actual configuration is job-specific.

WORKOVER



- rig / work-string well control
- Workover BOP system
- Tubular handling / barrier interface
- Choke / kill functionality as required by plan

Concept only.

Actual configuration is job-specific.

Confirm pressure / temperature rating, bore & tool compatibility, sealing interfaces, test criteria, shearability / non-shearables, and contingency function.



What Can Change the Barrier Envelope During Intervention?

The well may not change — but the operation can change which barrier elements are available, verified or exposed



Valve / flow path

OPENING A FLOW PATH

Opening tree / wellhead valves or shifting downhole flow-control devices.



Tool through closure plane

TOOL ACROLSS A BARRIER

Toolstring, wire, CT or pipe may prevent a barrier element from closing or sealing.



Pressure / thermal response

PRESSURE / THERMAL CHANGE

Pressure cycling, temperature changes and trapped pressure can alter barrier status.



Fluid / hydrostatic change

FLUID DISPLACEMENT

Losses, influx, gas migration or density changes can alter the hydrostatic barrier.



Leak / communication

INTEGRITY ANOMALY

Leak, communication or annulus response can invalidate an assumed barrier path.



Closure obstruction

STUCK / NON-SHEARABLES

A stuck tool or non-shearable item can reduce emergency closure options.

Barrier status is dynamic. Re-verify whenever the flow path, equipment or operating state changes.



Can We Safely Start?

Before the first valve is opened or the first tool is run, the team should be aligned on the well state, barriers and contingencies

01 **WELL STATUS**

- Current pressure / temperature and fluid state
- Recent trends, anomalies and annulus status



02 **BARRIER ENVELOPE**

- Current barrier schematic / defined WBEs
- Acceptance criteria and verification status



03 **PCE & INTERFACES**

- Correct configuration and ratings
- Function / pressure test and interface compatibility



04 **TOOLSTRING / ACCESS**

- Minimum ID, reach, deviation and restrictions
- Shearability / non-shearables and recovery plan



05 **OPERATING PLAN**

- Pressure limits, returns path and fluid handling
- Roles, communications, sequence and stop-work criteria



06 **CONTINGENCY**

- Loss of barrier / leak / stuck-tool response
- Emergency shutdown and escalation path



If barrier status, pressure-control capability or contingency response is unclear, the job is not ready to start



DISCUSSION #2

Can We Proceed With the Intervention?

The equipment is available. The objective is clear. But is the barrier picture clear enough to approve the job?

CASE FACTS



- Live gas well; planned slickline intervention



- Tubing pressure stable at surface



- A-annulus pressure observed; trend and source not yet confirmed



- Completion schematic available, but current barrier diagram is not updated



- Wireline PCE is available; job-specific configuration / test basis still to be confirmed

YOUR APPROVAL QUESTIONS

1. What is the current primary and secondary barrier envelope?



2. What could the annulus pressure indicate, and what must be verified first?



3. Does the planned PCE preserve the required barrier function through every step?



4. Are pressure ratings, interfaces and toolstring / shearability limitations understood?



5. What is the contingency if a barrier is lost or the tool cannot be recovered?



Would you approve, hold, or reject the job — and what evidence would change your decision?

5 MIN

From an Objective to an Executable Intervention

A good intervention program converts the technical objective into a controlled sequence with limits, hold points and contingencies

PROGRAM LOGIC



OBJECTIVE



BASIS & DATA



SEQUENCE



LIMITS & HOLD POINTS



CONTINGENCY

THREE PRACTITIONER PRINCIPLES

- The program must describe what we will do — and what will make us stop.
- Critical steps need verification and decision points, not assumptions.
- Contingencies must be executable with equipment, people and barriers actually available.

A program is a decision framework — not just a sequence of steps



What Must an Intervention Program Contain?

Enough detail to execute safely and consistently — without turning the program into an unreadable manual.



01 OBJECTIVE & SUCCESS CRITERIA

- What problem are we solving?
- What proves the job is successful?



02 WELL / DATA BASIS

- Current well status and key assumptions
- Completion, pressures, fluids, integrity, restrictions



03 BARRIER & WELL CONTROL

- Barrier envelope by phase
- PCE configuration, tests, limits and contingencies



04 EXECUTION SEQUENCE

- Step-by-step operations
- Hold points, verification and decision gates



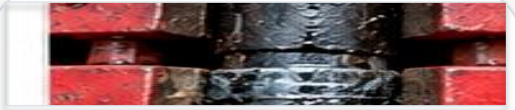
05 OPERATING ENVELOPE

- Pressure / load / rate / depth / returns limits
- Stop criteria and escalation thresholds



06 RESOURCES & INTERFACES

- People, services, responsibilities, SIMOPS
- Logistics, permits and communication



07 CONTINGENCIES

- Credible failure modes
- Immediate response and recovery path



08 HANDOVER & CLOSEOUT

- Restore / verify well status
- Data capture, lessons and final acceptance

Program detail should be proportional to risk and complexity — but critical barrier, limit and contingency information must never be implicit



Sequence the Job Around Verification and Decision Points

The operation should not simply move from Step 1 to Step 2; critical steps need a defined condition before proceeding.



PREPARE
Confirm data, program, roles & permits



RIG UP
Install equipment & interfaces



VERIFY
Function / pressure test and barrier status



ENTER WELL
Establish controlled access



EXECUTE
Perform intervention objective



RECOVER
Retrieve / reverse sequence



HANDOVER
Verify final well condition

EXAMPLE HOLD POINTS

	HP-1	**Before opening the well**	Barrier and PCE verification complete
	HP-2	**Before entering restriction / critical depth**	Toolstring, depth correlation and limits confirmed
	HP-3	**Before changing flow path / pumping**	Pressure limits, returns path and fluid plan confirmed
	HP-4	**Before restoring production**	Final well status / barriers / equipment confirmed

A hold point only works if the acceptance criterion and decision authority are clear



Define the Limits Before the Job Starts

The team should know what is normal, what requires attention, and what requires stopping the operation.

TYPICAL OPERATING PARAMETERS

	MONITOR / NORMAL BASIS	ATTENTION / LIMIT CUE
 **PRESSURE**	WHP / annulus / PCE / pumping pressure	pressure limit, unexpected trend or communication
 **LOAD / FORCE**	line tension, injector load, pipe force	overpull, compression, pipe-light / heavy transition
 **RATE / RETURNS**	pump rate, return rate, losses / gains	loss of returns, unexpected influx or solids
 **DEPTH / POSITION**	tool depth, correlation, restriction	depth mismatch or unexpected obstruction
 **FLUID / VOLUME**	density, concentration, pumped / returned volume	wrong fluid, volume imbalance, contamination
 **EQUIPMENT STATUS**	seal / BOP / pump / power / control system	loss of function or test validity

ACTION ZONES

GREEN — **Within approved envelope**

Continue & monitor

AMBER — **Approaching limit / unexplained trend**

Hold, assess, verify before continuing

RED — **Limit exceeded / barrier or control concern**

Stop, secure well, execute contingency

Stop criteria must be measurable or clearly observable



Most Intervention Failures Are Not Purely Technical

Execution depends on clear interfaces between the well owner, intervention crew, production, service companies and site control

WHO HAS AUTHORITY TO...

- open / close the well?
- change the program?
- accept a barrier test?
- stop the operation?

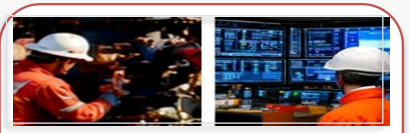


WELL OWNER / ENGINEERING



SITE / FACILITY CONTROL

INTERVENTION CONTROL



HSE / PERMIT / SIMOPS



PRODUCTION / OPERATIONS



INTERVENTION SUPERVISOR



SERVICE COMPANIES

SIMOPS must define incompatible activities, communication routes, control-room / site interfaces, and who can suspend the work



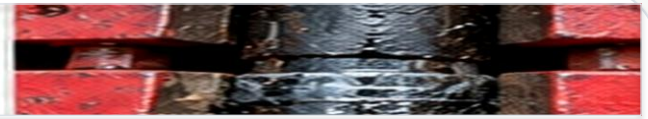
Plan the Recovery Before You Need It

A contingency is useful only if it identifies the trigger, immediate action, recovery path, decision authority and required resources



LOSS OF BARRIER

- Secure / isolate
- Verify remaining barrier
- Escalate per well-control contingency



STUCK TOOL / STRING

- Stop increasing exposure
- Confirm depth / forces
- Apply approved recovery sequence



NO / REDUCED RETURNS

- Stop / reduce pumping as required
- Check losses / leak path
- Reassess fluid / pressure strategy



UNEXPECTED PRESSURE

- Stop operation
- Stabilize / monitor
- Revalidate well state and barriers



PCE / EQUIPMENT FAILURE

- Secure with available barrier
- Isolate failed component
- Repair / replace before resume



OBJECTIVE NOT ACHIEVED

- Do not force the original method
- Re-diagnose
- Escalate / change method if justified

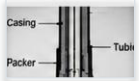


The contingency should reduce uncertainty — not introduce a second improvised operation

Would You Release This Program for Execution?

The objective and method are defined. Review the program basis and decide what must be closed before execution

PROGRAM SUMMARY

	**Objective**	Remove suspected tubing restriction
	**Method**	Coiled tubing cleanout
	**Well state**	Live well
	**PCE**	CT stack available
	**Sequence**	Rig up → pressure test → run CT → cleanout → pull out
	**Success criterion**	Restore access to target depth

WHAT IS STILL UNCLEAR?

-  Barrier envelope by phase is not shown
-  Maximum allowable pressure / load / pumping limits not stated
-  Returns route and loss-of-returns response are not defined
-  Restriction depth basis and cleanout endpoint are unclear
-  Hold points / stop criteria are not stated
-  Stuck CT / PCE failure contingencies are not executable from the program

RELEASE • HOLD • REJECT — What would you require before signing the program? **5 MIN**



When the Job Stops Looking Like the Program

Good intervention execution is not about forcing the original plan — it is about recognizing when the well is telling us something different

PRACTITIONER RESPONSE



RECOGNIZE



STABILIZE



VERIFY



DECIDE



RECOVER / ESCALATE

THE PRACTITIONER MINDSET

- Deviation is information — not merely delay.
- A safe pause is often cheaper than an uncontrolled recovery.
- Do not repeat a method until the failure mechanism is understood.

The moment conditions change, the decision basis must change too

A Deviation Is Data

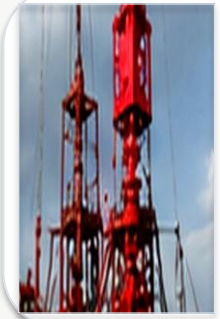
Do not normalize abnormal behavior just because the job is still moving forward



PRESSURE

Unexpected WHP / annulus / pumping trend

Potential barrier, restriction, fluid or flow-path change



LOAD / TENSION

Increasing overpull / drag / compression

Possible restriction, sticking, tool contact or geometry issue



RETURNS

Loss, gain, solids or gas behaviour changes

Possible loss zone, influx, communication or cleanup response



DEPTH / CORRELATION

Expected feature not found at expected depth

Potential depth-reference, movement or obstruction issue



TOOL RESPONSE

Tool does not function as expected

Mechanism, access, pressure differential or equipment issue



PRODUCTION RESPONSE

No improvement or different response than predicted

Diagnosis or intervention objective may be wrong

When two or more independent signals move together, treat the pattern as evidence — not coincidence

Stuck Tool? Do Not Start by Pulling Harder

A stuck or restricted toolstring is first a diagnosis problem: what changed, where, and under what load / pressure condition?

DIAGNOSTIC QUESTIONS



WHERE? Depth / correlation / last free point



WHAT CHANGED? Load, tension, pressure, circulation, tool response



WHAT MECHANISM? Restriction, debris, differential sticking, geometry, mechanical hang-up



WHAT IS SAFE? Approved loads, shear / release options, barrier state

RESPONSE LOGIC



STOP INCREASING EXPOSURE



STABILIZE WELL / LOADS



VERIFY DEPTH & MECHANISM



USE APPROVED RECOVERY



ESCALATE IF REQUIRED

Increasing force without understanding the mechanism can convert a recoverable restriction into a fishing job

Unexpected Pressure? Stabilize Before You Diagnose

An unexpected pressure response is a well-control signal until proven otherwise.



01 STABILIZE

- Stop the action causing further exposure
- Secure the well using available verified barriers
- Record pressures / rates / volumes and time



02 VERIFY

- Confirm gauges and measurement basis
- Check valve / flow-path configuration
- Compare tubing, annulus and pump / return behaviour



03 DECIDE

- Identify credible pressure source / communication path
- Revalidate barrier envelope
- Resume only under an approved revised basis

Do not explain pressure first and control it later. Control the exposure first; then investigate the cause.

The Intervention Worked — But the Well Did Not Improve

Technical execution can be successful while the intervention objective remains unmet.



WHAT WENT RIGHT OPERATIONALLY

- Tool reached planned depth
- Treatment / mechanical action completed as programmed
- No barrier or equipment failure
- Well returned to service safely



BUT THE OBJECTIVE WAS NOT MET

Do not immediately repeat the same method.

- Re-check the original diagnosis
- Confirm what the intervention actually changed
- Review new data / pressure / production response
- Decide whether the next step is repeat, change method, or stop

NEXT DECISION



REPEAT

same mechanism, execution gap identified



CHANGE METHOD

objective valid, capability insufficient



RE-DIAGNOSE

evidence does not support original cause



STOP / DEFER

risk, value or uncertainty no longer justified

When Should We Escalate to Workover?

Escalation is justified when the objective remains valid but lighter methods cannot safely or reliably achieve it.

ESCALATION TRIGGERS

	**ACCESS**	Required depth / completion component cannot be reached or manipulated
	**MECHANICAL CAPABILITY**	Need to pull / rerun tubing, packer or completion equipment
	**INTEGRITY**	Major repair / isolation requires rig-based capability
	**RECOVERY**	Fishing / stuck equipment exceeds rigless recovery capability
	**RECOMPLETION**	Objective requires zonal / completion architecture change
	**REPEAT FAILURE**	Rigless methods repeatedly fail without changing the failure mechanism

Escalation should be evidence-based — not driven by frustration with the previous method



What Would You Do Next?

Use the full decision logic from diagnosis, method selection, barriers and execution planning.

CASE FACTS



Oil producer declined from ~850 to ~320 BOPD



Tubing pressure gradually increased over the same period



A-annulus pressure now shows a small but repeatable increase after shut-in



Previous slickline gauge run stopped above target depth



CT is available next week; workover rig is available later



No recent downhole diagnostic confirms scale or tubing damage

YOUR DECISION

1. What is the leading diagnosis — and what evidence is still missing?



2. Would you run another diagnostic, intervene with CT, or wait for workover?



3. How does the annulus pressure affect the barrier strategy?



4. What would make you stop or change method during execution?



5. What result would justify escalation to workover?



Choose your next step — then defend the decision, barrier strategy and contingency.

8 MIN

Well Intervention in Practice — What Should Stay With You?

The equipment changes from job to job. The decision discipline should not.



01 START WITH THE WELL PROBLEM

Do not start with the available tool.



02 SYMPTOM ≠ DIAGNOSIS

Use history and data to narrow the cause.



03 OBJECTIVE BEFORE METHOD

Know what success must look like.



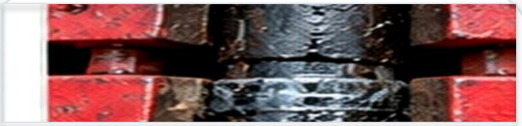
04 BARRIERS ARE DYNAMIC

Re-verify when the flow path or equipment changes.



05 DEFINE LIMITS BEFORE EXECUTION

The team must know when to hold and stop.



06 CONTINGENCIES MUST BE EXECUTABLE

Recovery needs people, equipment and authority.



07 DEVIATION IS INFORMATION

Stabilize, verify and decide — do not force the plan.



08 ESCALATE WITH EVIDENCE

Change method when capability, risk or recovery demands it.

The best intervention is not the most sophisticated one — it is the one that solves the right problem safely and deliberately

Thank You

Questions & Discussion

What should stay with us?

Solve the right problem.
Protect the well.
Execute deliberately. Escalate with evidence.



READ THE WELL
Problem • symptom • evidence



CONTROL THE EXPOSURE
Barriers • limits • well control



EXECUTE WITH DISCIPLINE
Method • contingency • escalation